

GAUSS'S LAW

Q.1. Define and explain electric flux?

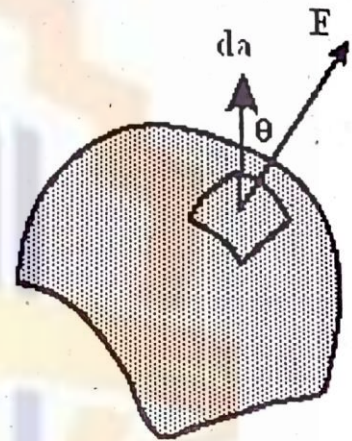
ELECTRIC FLUX

The total number of electric lines of force passing through certain area is called electric flux. It is denoted by ϕ_e . The scalar product of electric field $\vec{E}$ and vector area $\vec{da}$ of a surface S is equal to electric flux.

$$d\phi_e = \vec{E} \cdot \vec{da}$$

The net electric flux through entire surface is

$$\phi_e = \oint \vec{E} \cdot \vec{da}$$



Where $\vec{da}$ is called vector area whose magnitude is the area of the surface element and direction is outward drawn normal to surface. The circle on integral means closed surface.

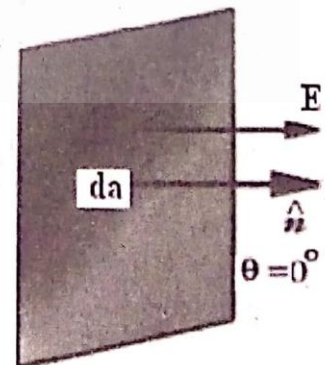
The SI unit of electric flux is called $\text{N-m}^2/\text{C}$. **MAXIMUM FLUX**

The electric flux through a surface will be maximum when surface is held perpendicular to electric field lines. The angle between normal $\hat{n}$ and electric field lines $\vec{E}$ is zero.

$$\phi_e = \oint \vec{E} \cdot \vec{da}$$

$$\phi_e = \oint E da \cos\theta$$

$$\phi_e = \oint E da \cos 0^\circ = \oint E da$$



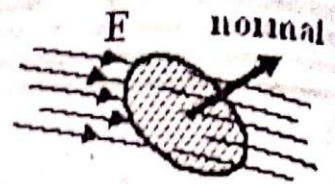
MINIMUM FLUX

When surface is held parallel to electric field lines, the angle between normal $\hat{n}$ and electric field lines is 90° . In this case, electric flux through surface is minimum.

$$\phi_e = \oint \vec{E} \cdot d\vec{a}$$

$$\phi_e = \oint E da \cos\theta$$

$$\phi_e = \oint E da \cos 90^\circ = 0$$

**Q. 2 State and prove Gauss's law and write its integral and differential form.****GAUSS'S LAW**

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

$$\phi_e = \frac{q}{\epsilon_0}$$

PROOF

Consider a $+q$ charge is placed in a closed surface. The electric field lines come out of the surface. Take small area element $d\vec{a} = da \hat{n}$ of the surface at point P. The normal $\hat{n}$ to area element makes angle θ with electric field line. The electric field at point P due to $+q$ charge having distance r is given as

$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

The electric flux through area element da is

$$d\phi_e = \vec{E} \cdot d\vec{a}$$

The net flux through entire surface is

$$\phi_e = \oint \vec{E} \cdot d\vec{a}$$

$$\phi_e = \oint \frac{kq}{r^2} \hat{r} \cdot da \hat{n}$$

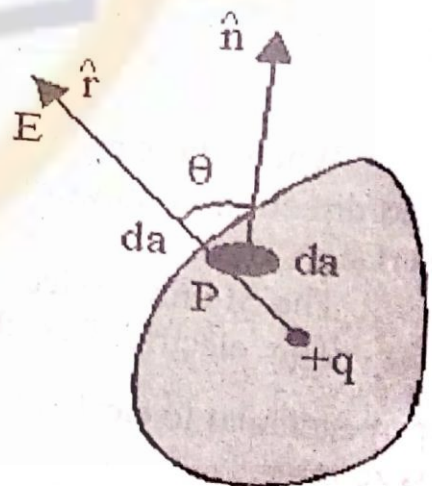
$$\phi_e = kq \oint \frac{da}{r^2} \hat{r} \cdot \hat{n} = kq \oint d\Omega$$

Where $\oint d\Omega$ is called total solid angle and its value is 4π

$$\phi_e = \frac{q(4\pi)}{4\pi\epsilon_0}$$

$$\phi_e = \frac{q}{\epsilon_0}$$

Hence, Gauss's law is proved.



INTEGRAL FORM

The volume charge density is defined as

$$\rho = \frac{dq}{dV}$$

$$dq = \rho dV$$

$$q = \int \rho dV$$

The net flux through entire surface is

$$\phi_e = \oint \vec{E} \cdot \vec{da} \quad \text{----- (1)}$$

Put eq(1) and eq(2) in Gauss's law

$$\phi_e = \frac{q}{\epsilon_0}$$

$$\oint \vec{E} \cdot \vec{da} = \frac{1}{\epsilon_0} \int_V \rho dV$$

This is called integral form of Gauss's law.

DIFFERENTIAL FORM

We know that

$$\oint \vec{E} \cdot \vec{da} = \frac{1}{\epsilon_0} \int_V \rho dV \quad \text{----- (3)}$$

Apply divergence theorem

$$\oint \vec{E} \cdot \vec{da} = \int (\vec{\nabla} \cdot \vec{E}) dV \quad \text{----- (4)}$$

Compare eq (1) and eq(2)

$$\int (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \int_V \rho dV$$

$$\int (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \int_V \rho dV$$

$$\int (\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0}) dV = 0$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

This is called differential form of Gauss's law

Q.3 State Gauss's law for electric field and derive Coulomb's law from Gauss's law? Using integral form of Gauss's law derive a formula to find intensity of electric field at a point due to source charge q .

GAUSS'S LAW

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

DEDUCTION OF COULOMB'S LAW FROM GAUSS'S LAW

Consider a $+q$ charge is placed at the center of sphere having radius r . The sphere is called gaussian surface. The electric lines of force are outward and perpendicular to the surface of the sphere.

Take small area element $\vec{da}$ of the sphere. The unit vector $\hat{n}$ indicates the direction of da and perpendicular to surface element. The angle between $\hat{n}$ and $\vec{E}$ is zero.

The flux through surface of sphere is

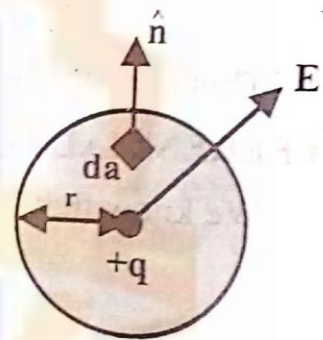
$$\phi_c = \oint \vec{E} \cdot \vec{da}$$

$$\phi_c = \oint E da \cos\theta$$

$$\phi_c = \oint E da \cos 0^\circ$$

$$\phi_c = \oint E da$$

$$\phi_c = E(4\pi r^2) \quad \text{----- (1)}$$



The Gauss's law states that electric flux passing through surface of sphere is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by the surface of sphere.

$$\phi_c = \frac{q}{\epsilon_0} \quad \text{----- (2)}$$

Comparing eq(1) and eq(2)

$$E(4\pi r^2) = \frac{q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{kq}{r^2}$$

$$E = \frac{kq}{r^2} \quad \text{----- (3)}$$

The electric force per unit charge is called electric field when a charge q_0 is placed in this field.

$$E = \frac{F_e}{q_0} \quad \text{----- (4)}$$

Comparing eq(3) and eq(4)

$$\frac{F_e}{q_0} = \frac{k q}{r^2}$$

$$F_e = \frac{k q q_0}{r^2}$$

This is statement of Coulomb's law. Hence, Coulomb's law is deduced from Gauss's law.

Q.4 Calculate electric field due to infinite line of charge by using Gauss's law.

ELECTRIC FIELD DUE TO INFINITE LINE OF CHARGE

Consider a straight positively charged wire having infinite length and negligible thickness. Now take a cylinder called gaussian surface having radius r and height h which encloses the some part of charged wire as shown in fig.

The P is the point on the surface of the cylinder where we want to evaluate electric field. Take a small length element dz of the line having charge dq . The linear charge density λ is given as

$$\lambda = \frac{dq}{dz}$$

$$dq = \lambda dz$$

The net charge enclosed by the cylinder having height h is

$$q = \lambda \int dz$$

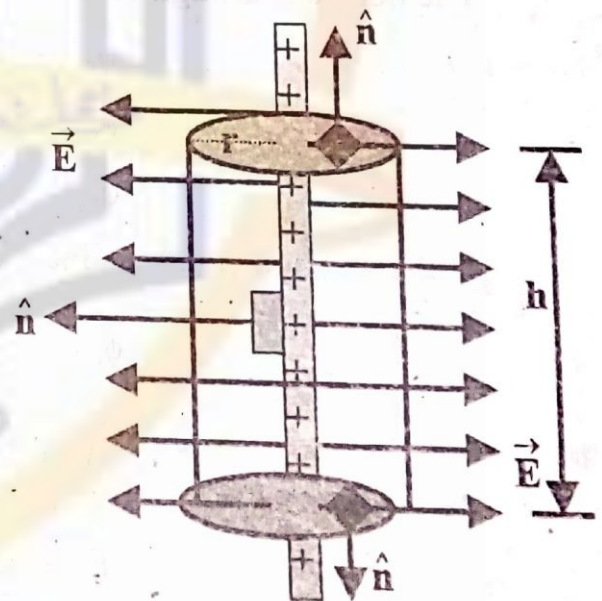
$$q = \lambda h \quad \text{----- (1)}$$

The cylinder has three surfaces

- ① Top circular surface
- ② Bottom circular surface
- ③ Curved surface

The electric field lines will be along radius of cylinder. Take a small area element da of each surface of the cylinder. The angle between normal $\hat{n}$ to area element of curved surface and electric field lines is zero.

The angle between normal $\hat{n}$ to area element of top circular surface and bottom circular surface and field lines is 90° .



Electric flux through top circular surface is

$$\phi_1 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_1 = \int E da \cos\theta$$

$$\phi_1 = \int E da \cos 90^\circ = 0$$

Electric flux through bottom circular surface is

$$\phi_2 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_2 = \int E da \cos\theta$$

$$\phi_2 = \int E da \cos 90^\circ = 0$$

Electric flux through curved surface is

$$\phi_3 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_3 = \int E da \cos\theta$$

$$\phi_3 = \int E da \cos 0^\circ = \int E da$$

$$\phi_3 = E (2\pi r h)$$

$$\text{Area} = (2\pi r)h$$

$$2\pi r$$

The net flux through entire surface of cylinder is,

$$\phi_e = \phi_1 + \phi_2 + \phi_3$$

$$\phi_e = 0 + 0 + E(2\pi r h)$$

$$\phi_e = E (2\pi r h) \quad \text{----- (2)}$$

Put eq(1) and eq(2) in Gauss's law

$$\phi_e = \frac{q}{\epsilon_0}$$

$$E(2\pi r h) = \frac{\lambda h}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

In vector form

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

Q.5 State Gauss's law and apply it to calculate electric field near an infinite charged sheet of charge?

GAUSS'S LAW

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

ELECTRIC FIELD DUE TO INFINITE SHEET OF CHARGE

Consider a thin positively charged sheet having infinite length and non-conducting surface. We have to calculate electric field $\vec{E}$ at points near the sheet by using Gauss's law.

Imagine a cylinder called gaussian surface having cross-sectional area A . Now pass this cylinder through sheet. Take small area element da of sheet having charge dq . The surface charge density σ is defined as

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

The charge of sheet enclosed by the cross-sectional area A of cylinder is

$$q = \int \sigma da$$

$$q = \sigma A \quad \text{----- (1)}$$

The gaussian cylinder has three surfaces

- ① Top capped surface
- ② Bottom capped surface
- ③ Curved surface.

The electric field $\vec{E}$ is parallel to end capped surfaces. The end-capped surfaces are at equal distance from the sheet, so electric field E has the same magnitude at the end caps.

The cylinder has three surfaces so flux through each surface is given as

Electric flux through top-capped surface is

$$\phi_1 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_1 = \int E da \cos\theta$$

$$\phi_1 = \int E da \cos 0^\circ = E A$$

Electric flux through bottom-capped surface is

$$\phi_2 = \int \vec{E} \cdot d\vec{a}$$

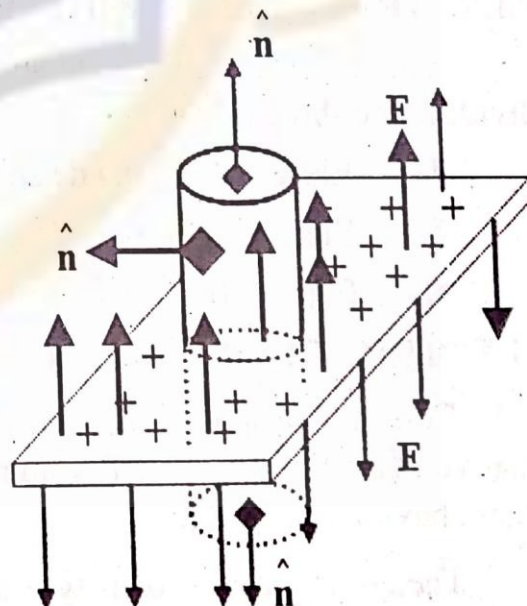
$$\phi_2 = \int E da \cos\theta$$

$$\phi_2 = \int E da \cos 0^\circ = E A$$

Electric flux through curved surface is

$$\phi_3 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_3 = \int E da \cos\theta$$



$$\phi_3 = \int E da \cos 90^\circ = 0$$

The net flux through entire surface of cylinder is

$$\phi_e = \phi_1 + \phi_2 + \phi_3$$

$$\phi_e = 0 + EA + EA$$

$$\phi_e = 2EA$$

(2)

Put eq (1) and eq (2) in Gauss's law is,

$$\phi_e = \frac{q}{\epsilon_0}$$

$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

In vector form

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{r}$$

Q.6 State Gauss's law and apply it to calculate electric field intensity at a point due to spherical volume charge distribution.

GAUSS'S LAW

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

ELECTRIC FIELD DUE TO SPHERICAL CHARGE DISTRIBUTION

The electric field due to spherical volume charge distribution using Gauss's law is divided into three parts

- ① Electric field inside spherical volume charge distribution
- ② Electric field outside spherical volume charge distribution
- ③ Electric on surface of spherical volume charge distribution.

ELECTRIC FIELD OUTSIDE SPHERICAL CHARGE DISTRIBUTION

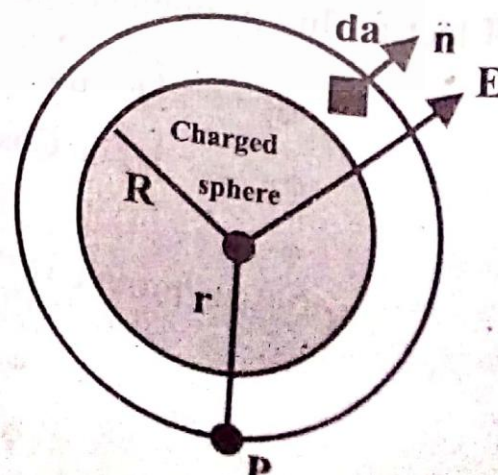
Consider a sphere of radius R having uniform positive charge. Take a small volume element dV of the sphere having charge dq .

The volume charge density is defined as

$$\rho = \frac{dq}{dV}$$

$$dq = \rho dV$$

The net charge enclosed in sphere is



$$q = \int \rho \, dV$$

$$q = \frac{4}{3} \pi R^3 \rho \quad \text{----- (1)}$$

Now take a point P at distance r from the center of charged sphere where we want to evaluate electric field. Imagine a gaussian surface in the form of sphere such that P lies on its surface and encloses charged sphere. Take small area element da of this gaussian sphere. The angle between radial electric field $\vec{E}$ and normal $\hat{n}$ to area element da is zero

Electric flux through area element da is

$$d\phi_e = \vec{E} \cdot \vec{da}$$

$$\phi_e = \int E \, da \cos\theta = \int E \, da \cos 0^\circ$$

The net flux through gaussian sphere is

$$\phi_e = E (4\pi r^2) \quad \text{----- (2)}$$

Put eq(1) and eq(2) in Gauss's law

$$\phi_e = \frac{q}{\epsilon_0}$$

$$E (4\pi r^2) = \frac{4\pi R^3 \rho}{3\epsilon_0}$$

$$E = \frac{R^3 \rho}{3\epsilon_0 r^2}$$

The electric field in vector form outside the spherical volume charge distribution of radius R at point p having distance r from center of charged sphere is

$$\vec{E} = \frac{R^3 \rho}{3\epsilon_0 r^2} \hat{r}$$

ELECTRIC FIELD INSIDE SPHERICAL CHARGE DISTRIBUTION

Consider a uniformly charged sphere having radius R . Take a small volume element dV of the sphere having charge dq . The volume charge density is defined as

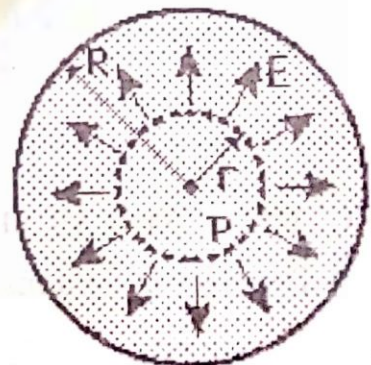
$$\rho = \frac{dq}{dV}$$

$$dq = \rho \, dV$$

The net charge enclosed in sphere is

$$q = \int \rho \, dV$$

$$q = \frac{4}{3} \pi R^3 \rho \quad \text{----- (1)}$$



Now take a point P inside charged sphere having distance r from the center of charged sphere where we want to evaluate electric field. Imagine a gaussian surface in the form of sphere such that P lies on its surface.

Charge enclosed in gaussian sphere of radius r is

$$q' = \frac{4}{3} \pi r^3 \rho \quad \text{----- (2)}$$

Divide eq (2) and eq(1)

$$\frac{q'}{q} = \frac{\frac{4}{3} \pi R^3 \rho}{\frac{4}{3} \pi r^3 \rho}$$

$$q' = \frac{q r^3}{R^3} \quad \text{----- (3)}$$

Now consider small area element da of gaussian sphere. The angle between electric field $\vec{E}$ and normal $\hat{n}$ to area element da is zero

Electric flux through area element da is

$$\phi_e = \oint \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int E da \cos \theta = \int E da \cos 0^\circ$$

The net flux is

$$\phi_e = E (4\pi r^2) \quad \text{----- (4)}$$

Put eq(3) and eq(4) in Gauss's law

$$\phi_e = \frac{q'}{\epsilon_0}$$

$$E (4\pi r^2) = \frac{q r^3}{R^3 \epsilon_0}$$

$$E = \frac{q r}{4\pi \epsilon_0 R^3}$$

ELECTRIC FIELD AT SURFACE OF SPHERE

The electric field inside charge sphere is

$$E = \frac{q r}{4\pi \epsilon_0 R^3}$$

Put $r = R$ to calculate electric field at the surface of sphere

$$E = \frac{q R}{4\pi \epsilon_0 R^3} = \frac{kq}{R^2}$$

Q. 7 State Gauss's law and apply it to calculate electric field intensity due to spherical charged shell of constant surface charge density.

GAUSS'S LAW

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

SHELL THEOREMS

There are two shell theorems established by using Gauss's law for uniform spherical shell of charge having constant surface charge density and stated as

1 FIRST SHELL THEOREM

A uniform spherical shell of charge behaves for external points in such a way that all its charge was concentrated at its center.

PROOF

Consider a positively charged spherical shell such as charge sprayed uniformly over the surface of balloon having radius R . Take a small area element da of shell having charge dq .

The surface charge density σ is defined as

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

The net charge on the shell is

$$q = \sigma (4\pi R^2) \quad \text{----- (1)}$$

Consider a point P outside shell having distance r from the center of spherical charged shell where we want to evaluate electric field. Imagine a gaussian spherical shell which encloses charged shell such that P lies on its surface.

Now take a small area element da of gaussian shell having radius r . The angle between normal $\hat{n}$ to this area element and electric field $\vec{E}$ is zero.

Electric flux through area element da is

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int E da \cos\theta = \int E da \cos 0^\circ$$

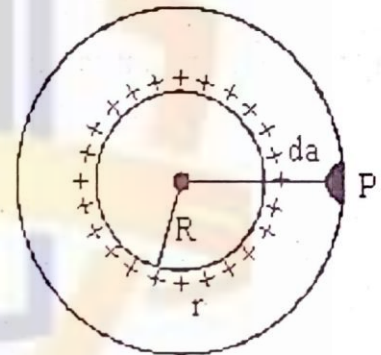
The net flux is

$$\phi_e = E (4\pi r^2) \quad \text{----- (2)}$$

Put eq (2) in Gauss's law.

$$\phi_e = \frac{q}{\epsilon_0}$$

$$E (4\pi r^2) = \frac{q}{\epsilon_0}$$



$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{kq}{r^2}$$

This is electric field at point P due to point charge q. It means all the charge on the surface of hollow spherical shell is concentrated at its center. Hence, first shell theorem is proved.

② SECOND SHELL THEOREM

A uniform charged spherical shell exerts no electrostatic force on charged particle placed inside the shell

PROOF

Consider a positively charged spherical shell having radius R and uniform surface charge density σ . The net charge on shell is q

Take a point P inside the shell where we want to evaluate electric field. The point P has distance r from center of spherical charged shell. Imagine a gaussian spherical shell which encloses no charge such that P lies on its surface.

Now consider small area element da of gaussian shell having radius r. The angle between normal $\hat{n}$ to this area element and electric field $\vec{E}$ is zero.

Electric flux through area element da is

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int E da \cos\theta$$

$$\phi_e = \int E da \cos 0^\circ$$

The total flux is

$$\phi_e = E (4\pi r^2) \quad \text{----- (2)}$$

Put eq (2) in Gauss's law

$$\phi_e = \frac{q}{\epsilon_0}$$

$$E (4\pi r^2) = \frac{q}{\epsilon_0}$$

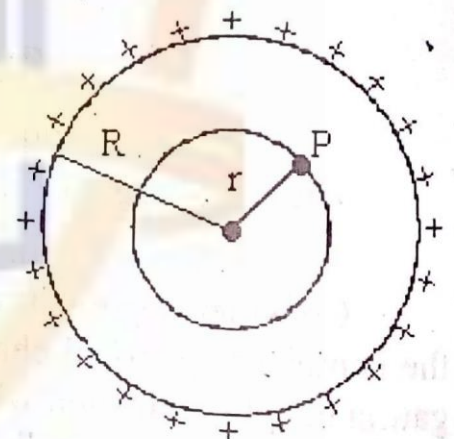
Inside the Gaussian surface, charge is zero ($q = 0$)

$$E = \frac{0}{4\pi\epsilon_0 r^2} = 0$$

The electric force on charge placed inside gaussian shell.

$$F_e = qE = q(0) = 0$$

It means a uniform charged spherical shell exerts no electrostatic force on charged particle placed inside the shell. Hence, second shell theorem is proved



Q.8 Show that by using Gauss's law that an excess charge placed on an isolated conductor moves entirely to the outer surface of the conductor and none of the excess charge is found within the body of the conductor.

CHARGED ISOLATED CONDUCTOR

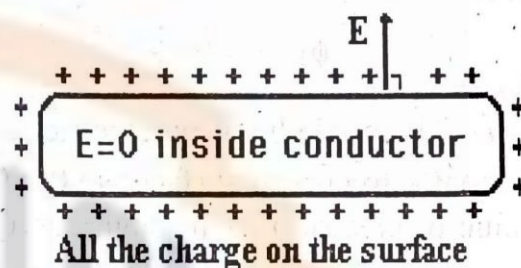
Consider a cross-section of an isolated conductor which is charged positively as shown in fig. Imagine a gaussian surface that lies inside the surface of the conductor. If we prove that electric field inside the conductor is zero, it means all of its charge is on its outer surface.

Suppose that electric field is present inside the isolated conductor. The electric field will exert force on electrons and current is established in conductor, but it is experimentally proved that isolated conductor has no current.

If electric field inside isolated charged conductor is zero. The value of electric flux is

$$\phi_e = E A$$

$$\phi_e = 0$$



It means by Gauss's law electric flux through gaussian surface which is inside isolated charged conductor must be zero.

The Gauss's law is

$$\phi_e = \frac{q}{\epsilon_0}$$

$$0 = \frac{q}{\epsilon_0}$$

It means net charge $q = 0$. If the added charge is not inside the gaussian surface, it can be outside the surface. The gaussian surface is very close to outer surface of the conductor as shown in fig. Hence, charge added is on outer surface of the conductor.

It is concluded that an excess charge placed on an isolated conductor moves entirely to the outer surface of the conductor and none of the excess charge is found within the body of the conductor.

Q.9 Calculate electric field outside the charged isolated conductor by using Gauss's law?

EXTERNAL ELECTRIC FIELD

Consider a positively charged isolated conductor. The excess charge placed on an isolated conductor moves entirely to the outer surface of the conductor. Imagine a gaussian surface in the shape of cylinder.

The cylinder has three portions. The end caps are parallel to the surface of the conductor and one end capped is inside and other is outside the surface as shown in fig. The walls of cylinder are perpendicular to surface. The area of each end capped surface is A .

The electric field just outside a charged isolated conductor is at right angle to the surface of conductor. The value of electric flux through end capped outside surface is

$$\phi_1 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_1 = \int E da \cos\theta = \int E da \cos 0^\circ$$

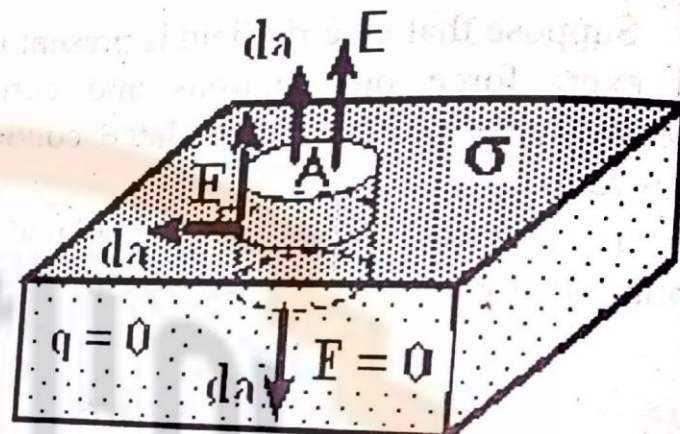
$$\phi_1 = EA$$

The electric field inside a charged isolated conductor is zero. The value of electric flux through end cap inside surface is

$$\phi_2 = \int \vec{E} \cdot d\vec{a} = \int E da \cos\theta$$

$$\phi_2 = \int (0) da \cos\theta$$

$$\phi_2 = 0$$



The angle between electric field $\vec{E}$ and normal $\hat{n}$ to curved surface is 90° . Therefore, value of electric flux through curved surface is

$$\phi_3 = \int \vec{E} \cdot d\vec{a}$$

$$\phi_3 = \int E da \cos\theta = \int E da \cos 90^\circ$$

$$\phi_3 = 0$$

The net flux is

$$\phi_e = \phi_1 + \phi_2 + \phi_3$$

$$\phi_e = EA + 0 + 0$$

$$\phi_e = EA$$

----- (1)

If dq charge is enclosed by da area of gaussian surface. The surface charge density σ is defined as

$$\sigma = \frac{dq}{da}$$

$$q = \int \sigma da$$

The net charge enclosed by end capped which is outside having area A is

$$q = \sigma A$$

----- (2)

The Gauss's law is

$$\phi_e = \frac{q}{\epsilon_0}$$

$$E A = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

In vector form

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{r}$$

This is electric field just outside the charged isolated conductor.

SHORT QUESTIONS

Q:1 Suppose that an electric field in some region is found to have a constant direction but to be decreasing in strength in that direction. What do you conclude about the charge in the region?

Ans: The electric field and charge are related as $E = \frac{kq}{r^2}$. It shows that strength of electric field decreases when strength of charge decreases. The direction of electric field remains constant.

Q:2 Is it precisely true that Gauss's law states that the total number of lines of force crossing any closed surface in the outward direction is proportional to the net positive charge enclosed within the surface.

Ans: The Gauss's law states that total electric flux is equal to $1/\epsilon_0$ times total charge enclosed by that surface.

$$\phi_e = \frac{q}{\epsilon_0}$$

$$\phi_e \propto q$$

Therefore, it is precisely true that Gauss's law states that the total number of lines of force crossing any closed surface in the outward direction called electric flux is proportional to the net positive charge enclosed within the surface.

Q:3 A point charge is placed at the center of a spherical gaussian surface. Is flux changed (a) if the surface is replaced by a cube of the same volume. (b) If the sphere is replaced by a cube of one tenth the volume (c) If the charge is moved off center in the original sphere, still remaining inside (d) If the charge is moved just outside the original sphere (e) If a second charge is placed near and outside the original sphere and (f) If a second charge is placed inside the gaussian surface.

Ans: (a). A point charge is placed at the center of a spherical gaussian surface. If the surface is replaced by a cube of the same volume, the flux will not change.
(b). A point charge is placed at the center of a spherical gaussian surface. If the surface is replaced by a cube of one-tenth the volume, the flux will not change.

(c). A point charge is placed at the center of a spherical gaussian surface. If the charge is moved off center in the original sphere, still remaining inside, the flux will not change.

(d). A point charge is placed at the center of a spherical gaussian surface. If the charge is moved just outside the original sphere, the flux will change.

(e). A point charge is placed at the center of a spherical gaussian surface. If a second charge is placed near and outside the original sphere, the flux will not change.

(f). A point charge is placed at the center of a spherical gaussian surface. If a second charge is placed inside the gaussian surface, the flux will change.

Q:4 In Gauss's law $\epsilon_0 \oint \vec{E} \cdot d\vec{a} = q$. Is $\vec{E}$ necessarily the electric field attributable to the charge q .

Ans: The electric field and charge are related as $E = kq/r^2$. The term $\vec{E}$ necessarily the electric field attributes to the charge q in Gauss's law $\epsilon_0 \oint \vec{E} \cdot d\vec{a} = q$.

Q:5 A surface encloses an electric dipole. What can you say about electric flux for this surface?

Ans: The pair of positive charge and negative charge of same magnitude having constant distance is called electric dipole. The electric flux through a surface will be zero when a surface encloses an electric dipole because inward flux is balanced by outward flux.

$$\phi = \frac{(+q) + (-q)}{2} = 0$$

Q:6 Is E necessarily zero inside a charged rubber balloon if the balloon is (a) spherical (b) sausage shaped. For each shape, assume that charge to be distributed uniformly over the surface.

Ans: The charge is distributed over the balloon surface. There is no charge inside the balloon so $q = 0$

The Gauss's law states that total electric flux is equal to $1/\epsilon_0$ times total charge enclosed by that surface.

$$\phi_c = \frac{q}{\epsilon_0} = \frac{0}{\epsilon_0}$$

$$\phi_c = 0$$

$$EA = 0$$

It means $E = 0$ because A cannot be zero. It proves E is necessarily zero inside a charged rubber balloon if the balloon is (a) spherical or (b) sausage shaped.

Q:7 A surface encloses two equal and opposite charges. What can you say about electric flux for the surface?

Ans: The Gauss's law states that total electric flux is equal to $1/\epsilon_0$ times total charge enclosed by that surface. The net charge is zero in a surface which encloses two equal and opposite charges. Therefore, electric flux for the surface is zero.

Q:8 Discuss the positive, negative and zero flux passing through a closed surface with reference to "sources" and "sinks".

Ans: The Gauss's law states that total electric flux is equal to $1/\epsilon_0$ times total charge enclosed by that surface. The electric flux is positive when surface encloses $+q$ charge (sources). The electric flux is negative when surface encloses $-q$ charge (sinks). The electric flux is zero when surface encloses $+q$ and $-q$ charge.

Q:9 How do the gravitational and electric forces resemble each other. How do they differ?

Ans: The similarities between gravitational force and electric force are

- 1 The electric force and gravitational force is conservative.
- 2 The electric force and gravitational force vary inversely with the square of the distance between two charges or two masses.

The differences between gravitational force and electric force are

- 1 The electric force is medium dependent but gravitational force is independent of medium effects.
- 2 The gravitational force is weaker as compared to electric force.
- 3 The electric force is attractive and repulsive but gravitational force is only attractive.

Q:10 What will be value of electric flux if surface does not enclose any charge.

Ans: The Gauss's law states that total electric flux is equal to $1/\epsilon_0$ times total charge enclosed by that surface. The electric flux will be zero when surface does not enclose any charge

Q:11 How will you calculate the total charge on a spherical body if its radius R and the surface charge density σ are given.

Ans. The surface charge density is defined as

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

$$q = \int \sigma da = \sigma (4\pi R^2)$$

Q:12 Define gaussian surface?

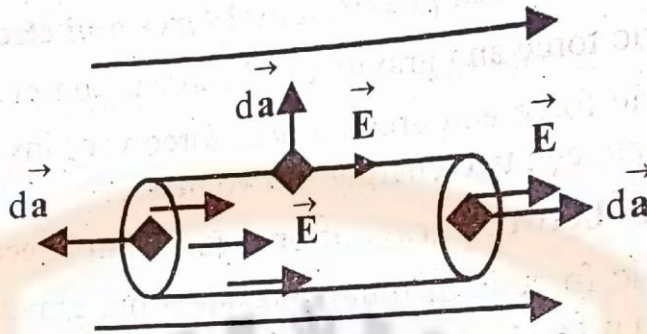
Ans. The power of Gauss's law in evaluating electric field is that it applies to any surface. It is often convenient to construct an imaginary surface called a gaussian surface to take advantage of the symmetry of the physical situation.

SAMPLE PROBLEMS

P.1 A hypothetical closed cylinder of radius R is immersed in a uniform electric field E , the cylinder axis being parallel to the field. Find electric flux for this closed surface.

Sol.

Consider a hypothetical closed cylinder of radius R which is immersed in a uniform electric field E such that cylinder axis being parallel to the electric field as shown in fig.



The cylinder has three surfaces called left capped surface, right capped surface and curved surface. The net flux through surface of cylinder is

$$\phi = \phi_1 + \phi_2 + \phi_3$$

$$\phi = \int E \, da \cos 180^\circ + \int E \, da \cos 90^\circ + \int E \, da \cos 0^\circ$$

$$\phi = -EA + 0 + EA$$

$$\phi = 0$$

P.2 The electric field just above the surface of the charged drum of photocopying machine has magnitude of electric field $2.3 \times 10^5 \text{ N/C}$. What is the surface charge density on the drum if it is conductor?

Sol.

DATA

$$E = 2.3 \times 10^5 \text{ N/C}$$

$$\sigma = ?$$

CALCULATIONS

$$E = \frac{\sigma}{\epsilon_0}$$

$$\sigma = \epsilon_0 E$$

$$\sigma = 2.3 \times 10^5 \times 8.85 \times 10^{-12}$$

$$\sigma = 2.03 \times 10^{-6} \text{ C/m}^2$$

P.3 The magnitude of electric field normally present in earth atmosphere just above the surface of earth is 150 N/C directed downward. What is net surface charge density assume that earth is a conductor?

Sol.

DATA

$$E = 150 \text{ N/C}$$

$$\sigma = ?$$

CALCULATIONS

The electric field at point due to conductor is

$$E = \frac{\sigma}{\epsilon_0}$$

$$\sigma = \epsilon_0 E = 8.85 \times 10^{-12} \times 150$$

$$\sigma = 1.3 \times 10^{-9} \text{ C/m}^2$$

The σ will be -ve because electric field is directed downward or into earth

$$q = \sigma A = \sigma(4\pi R^2)$$

$$q = 1.33 \times 10^{-9} \times 4\pi(6.4 \times 10^6)^2$$

$$q = +6.8 \times 10^5 \text{ C}$$

24. A plastic rod whose length is 220 cm and whose radius is 3.6 mm, carries a negative charge of magnitude $3.8 \times 10^{-7} \text{ C}$ spread uniformly over its surface. What is the electric field near the mid-point of rod at a point of its surface?

Sol.

DATA

$$r = 3.6 \times 10^{-3} \text{ m}$$

$$L = 220 \times 10^{-2} \text{ m}$$

$$q = -3.8 \times 10^{-7} \text{ C}$$

$$E = ?$$

CALCULATIONS

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{q}{2\pi\epsilon_0 r L} \quad \left[\lambda = \frac{q}{L} \right]$$

$$E = \frac{-3.8 \times 10^{-7}}{2\pi \times 8.85 \times 10^{-12} \times 3.6 \times 10^{-3} \times 220 \times 10^{-2}}$$

$$E = -8.6 \times 10^5 \text{ N/C}$$

EXERCISE PROBLEMS

21. A plastic rod of radius 2.73 mm has a linear charge density $\lambda = 2.73 \mu\text{C/m}$. What is the electric field near the mid point of the rod at a point on its surface.

Sol.

DATA

$$r = 2.73 \text{ mm} = 2.73 \times 10^{-3} \text{ m}$$

$$\lambda = 2.73 \mu\text{C/m} = 2.73 \times 10^{-6} \text{ C/m}$$

$$E = ?$$

CALCULATIONS

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$E = \frac{2.73 \times 10^{-6}}{2\pi \times 8.85 \times 10^{-12} \times 2.73 \times 10^{-3}} = 1.7 \times 10^7 \text{ N/C}$$

- P.2** The square surface measure 3.2mm on each side. It is immersed in uniform electric field $E = 1800 \text{ N/C}$. The field lines makes angle 65° with normal. Calculate flux.

Sol.

DATA

$$E = 1800 \text{ N/C}$$

$$\theta = 65^\circ$$

$$A = 3.2 \text{ mm}^2 = 10.2 \times 10^{-6} \text{ m}^2$$

CALCULATIONS

$$\phi_e = E A \cos\theta$$

$$\phi_e = 1800 \times 10.2 \times 10^{-6} \cos 65^\circ$$

$$\phi_e = 7.78 \times 10^{-3} \text{ N-m}^2/\text{C}$$

- P.3** A proton of mass m orbits with a speed $V = 284 \text{ km/s}$ just outside a charged sphere of radius $r = 2.13 \text{ cm}$ so that value of $mV^2/r = 2.87 \times 10^{-18} \text{ (SI unit)}$. Find charge on sphere?

Sol.

DATA

$$mV^2/r = 2.87 \times 10^{-18} \text{ (SI unit)}$$

$$V = 284 \text{ km/s} = 284 \times 10^3 \text{ m/s}$$

$$r = 2.13 \text{ cm}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$q = ?$$

CALCULATIONS

$$F = \frac{kqe}{r^2}$$

$$\text{and } F = \frac{mV^2}{r}$$

$$\frac{mV^2}{r} = \frac{kqe}{r^2}$$

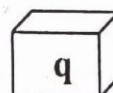
$$q = \frac{mV^2 r}{ke}$$

$$q = \frac{2.87 \times 10^{-18}}{9 \times 10^9 \times 1.6 \times 10^{-19}} = 1.9 \times 10^{-9} \text{ C}$$

P.4 A point charge q is placed at centre of a cube of edge a . What is the electric flux through each of the cube face.

Sol.

$$\phi_c = \frac{q}{\epsilon_0}$$



✓ P.5 A proton orbits with a speed 294 km/s just outside a charged sphere of radius 1.13 cm. Find charge on sphere?

Sol.

DATA

The charged sphere inside the orbit behaves as point charge having value q

Charge of sphere = $q = ?$

$$V = 294 \text{ km/s} = 294 \times 10^3 \text{ m/s}$$

$$r = 1.13 \text{ cm}$$

$$\text{Charge of proton} = e = 1.6 \times 10^{-19} \text{ C}$$

CALCULATIONS

$$F = \frac{kqe}{r^2} \quad \text{and} \quad F = \frac{mV^2}{r}$$

$$\frac{mV^2}{r} = \frac{kqe}{r^2}$$

$$q = \frac{mV^2}{ke} r$$

$$q = \frac{1.67 \times 10^{-27} \times (294 \times 10^3)^2 (1.13 \times 10^{-2})}{9 \times 10^9 \times 1.6 \times 10^{-19}}$$

$$q = +1.13 \times 10^{-9} \text{ C}$$

✓ P.6 An infinite line of charge produces a field of $4.52 \times 10^4 \text{ N/C}$ at a distance of 1.96 m. Calculate linear charge density.

Sol.

DATA

$$E = 4.52 \times 10^4 \text{ N/C}$$

$$r = 1.96 \text{ m}$$

$$\lambda = ?$$

CALCULATIONS

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\lambda = E(2\pi\epsilon_0 r)$$

$$\lambda = 4.52 \times 10^4 \times 2\pi (8.85 \times 10^{-12})(1.96)$$

$$\lambda = 4.9 \times 10^{-6} \text{ C/m}$$

P.7 A point charge of $1.84 \mu\text{C}$ is at the center of cubical gaussian surface 55 cm on edge. Find the flux ϕ_e through Surface?

Sol.

DATA

$$q = 1.84 \mu\text{C} = 1.84 \times 10^{-6} \text{ C}$$

$$a = 55 \times 10^{-2} \text{ m}$$

$$\phi_e = ?$$

CALCULATIONS

$$\phi_e = \frac{q}{\epsilon_0}$$

$$\phi_e = \frac{1.84 \times 10^{-6}}{8.85 \times 10^{-12}} = 207909 \text{ N-m}^2/\text{C}$$

P.8 A cube with 1.4 m edge is oriented as shown in fig. In a uniform region of electric field. Find the electric flux through the right face if electric field in N/C is given by (a) $6 \hat{i}$ (b) $-2 \hat{j}$ (c) $-3 \hat{i} + 4 \hat{k}$ (d) Calculate the total flux through the cube for each of these fields.

Sol.

$$\text{Area of right face} = (1.4)^2 = 1.96 \text{ m}^2 \hat{j}$$

$$(a) \vec{E} = 6 \hat{i}$$

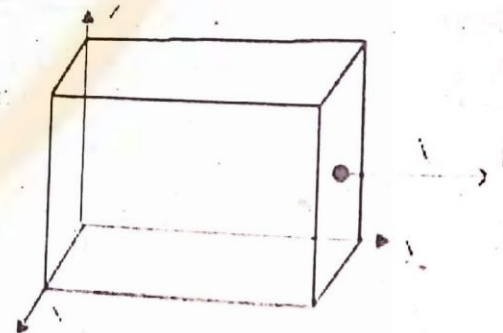
$$\phi_e = \vec{E} \cdot \vec{A} = (6 \hat{i}) \cdot (1.96) \hat{j} = 0$$

$$(b) \vec{E} = -2 \hat{j}$$

$$\phi_e = \vec{E} \cdot \vec{A} = (-2 \hat{j}) \cdot (1.96 \hat{j}) = -3.92 \text{ N-m}^2/\text{C}$$

$$(c) \vec{E} = -3 \hat{i} + 4 \hat{k}$$

$$\phi_e = \vec{E} \cdot \vec{A} = (-3 \hat{i} + 4 \hat{k}) \cdot (1.96 \hat{j}) = 0$$



- (d) Total flux through the cube when $\vec{E} = 6 \hat{i}$ is
 Flux through face perpendicular to y-axis is

$$\phi_1 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot 1.96 \hat{j} = 0$$

Flux through face perpendicular to -y-axis is

$$\phi_2 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot (-1.96 \hat{j}) = 0$$

Flux through face perpendicular to x-axis is

$$\phi_3 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot (1.96 \hat{i}) = 11.76 \text{ Nm}^2/\text{C}$$

Flux through face perpendicular -x-axis is

$$\phi_4 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot (-1.96 \hat{i}) = -11.76 \text{ Nm}^2/\text{C}$$

Flux through face perpendicular to z-axis is

$$\phi_5 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot (1.96 \hat{k}) = 0$$

Flux through face perpendicular to -z-axis is

$$\phi_6 = \vec{E} \cdot \vec{A} = 6 \hat{i} \cdot (-1.96 \hat{k}) = 0$$

Total flux

$$\phi_c = \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6$$

$$\phi_c = 0 + 0 + 11.76 - 11.76 + 0 + 0$$

$$\phi_c = 0$$

Similarly, for other fields we can find flux.

Calculate ϕ_e through (a) flat base (b) through the curved surface of a hemisphere of radius R , the field E is uniform and parallel to the axis of the hemisphere and the lines of field enter through flat base?

Sol.

The flux through the flat base of hemi sphere

$$\phi_1 = \vec{E} \cdot \vec{A}$$

$$\phi_1 = E A \cos 180^\circ = -E A$$

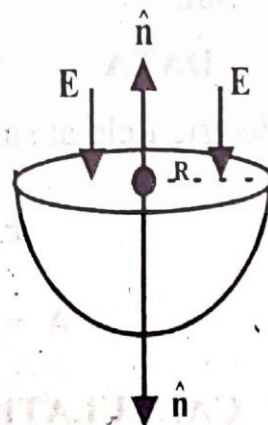
$$\phi_1 = -E(\pi R^2)$$

The flux through the curved surface is ($\theta = 0^\circ$)

$$\phi_2 = \vec{E} \cdot \vec{A}$$

$$\phi_2 = E A \cos 0^\circ = E A$$

$$\phi_2 = E(2\pi R^2)$$



P.10 The net electric flux through each face of a dice has magnitude in units of $10^3 \text{ N-m}^2/\text{C}$ equal to the number N spots on face (1 through 6). The flux is inward for N odd and outward for even. What is the net charge inside.

Sol.

DATA

The inward flux for N -odd spots of dice

$$\phi_1 = (1 + 3 + 5)10^3$$

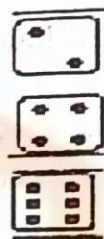
$$\phi_1 = 9 \times 10^3 \text{ N-m}^2/\text{C}$$



The outward flux for N -even spots of dice

$$\phi_2 = (2 + 4 + 6)10^3$$

$$\phi_2 = 12 \times 10^3 \text{ N-m}^2/\text{C}$$



CALCULATIONS

Net flux is

$$\phi_e = \phi_2 - \phi_1$$

$$\phi_e = 12 \times 10^3 - 9 \times 10^3$$

$$\phi_e = 3 \times 10^3 \text{ N-m}^2/\text{C}$$

Gauss's law is

$$\phi_e = \frac{q}{\epsilon_0}$$

$$3 \times 10^3 = \frac{q}{8.85 \times 10^{-12}}$$

$$q = 2.7 \times 10^{-8} \text{ C}$$

P.11 A point charge $+q$ is at a distance $d/2$ from a square surface of side d and is directly above the center of the square. Find the electric flux through square?

Sol.

DATA

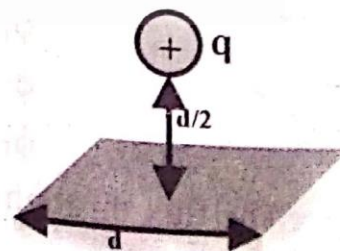
The electric field at surface of square (one face of cube) is

$$E = \frac{kq}{(d/2)^2} = \frac{4kq}{d^2}$$

$$A = d \times d = d^2$$

CALCULATIONS

$$\phi_e = \vec{E} \cdot \vec{A} = EA \cos 0^\circ$$



$$\phi_e = E A = \frac{4kq}{d^2} \cdot d^2$$

$$\phi_e = 4kq = \frac{4q}{4\pi\epsilon_0} = \frac{q}{\pi\epsilon_0}$$

P.12 A butter fly net is in uniform electric field E as shown in fig. The rim of circle of radius "a" is aligned perpendicular to field. Find the electric flux through the net relative to the outward normal.

Sol.

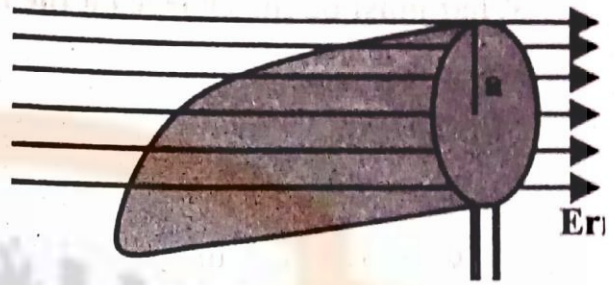
DATA

Uniform Electric Field = $\vec{E}$

Radius of rim = a

Area of rim = πa^2

$\theta = 0$



CALCULATIONS

$$\phi_e = \vec{E} \cdot \vec{A} = E A \cos 0^\circ$$

$$\phi_e = EA = E(\pi a^2)$$

P.13 A uniformly charged conducting sphere of 1.22m radius has surface charge density $8.13 \mu\text{C}/\text{m}^2$ (a) Find the charge on sphere (b) What is the electric flux leaving surface (c) Calculate electric field at the surface of sphere?

Sol.

DATA

$R = 1.22 \text{ m}$

$\sigma = 8.13 \times 10^{-6} \text{ C}/\text{m}^2$

(a) $q = ?$

(b) $\phi_e = ?$

(c) $E = ?$

CALCULATIONS

(a) $E = \frac{\sigma}{\epsilon_0} = \frac{8.13 \times 10^{-6}}{8.85 \times 10^{-12}} = 9.18 \times 10^5 \text{ N/C}$

$q = \sigma A$

$q = \sigma(4\pi R^2)$

$q = 8.13 \times 10^{-6} \times 4\pi \times (1.22)^2 = 1.5 \times 10^{-4} \text{ C}$

$\phi_e = EA = E(4\pi R^2) = 9.18 \times 10^5 \times 4\pi \times (1.22)^2$

(b)

$\phi_e = 1.7 \times 10^7 \text{ N-C}/\text{m}^2$

$$(c) \quad E = \frac{kq}{R^2}$$

$$E = \frac{9 \times 10^9 \times 1.5 \times 10^{-4}}{(1.22)^2} = 907014.2 \text{ N/C}$$

P.14 The drum of photocopying machine having length 42cm and a diameter of 12 cm with surface charge density $2.0 \times 10^{-6} \text{ C/m}^2$. (a) What is the total charge on the drum? (b) The manufacture wishes to produce a desktop version of the machine. This requires reducing the size of the drum to a length of 28 cm and a diameter of 8 cm. The electric field remains constant. What must be the charge on the new drum?

Sol.

DATA

$$\sigma = 2.0 \times 10^{-6} \text{ C/m}^2$$

$$d = 12 \times 10^{-2} \text{ m}$$

$$L = 42 \text{ cm}$$

$$q = ?$$

CALCULATIONS

$$\sigma = \frac{q}{A} = \frac{q}{2\pi rL}$$

$$q = 2 \times 10^{-6} \times 2\pi \times 6 \times 10^{-2} \times 42 \times 10^{-2}$$

$$q = 3.16 \times 10^{-7} \text{ C}$$

$$(b) \quad \sigma = 2.0 \times 10^{-6} \text{ C/m}^2$$

$$L = 28 \text{ cm}$$

$$d = 8 \text{ cm}$$

$$q = ?$$

CALCULATIONS

$$q = \sigma 2\pi rL$$

$$q = 2.0 \times 10^{-6} \times 2\pi \times 4 \times 10^{-2} \times 28 \times 10^{-2}$$

$$q = 1.4 \times 10^{-7} \text{ C}$$

P.15 An electron remains stationary in electric field directed downward in the earth gravitational field. If electric field is due to charge on two parallel conducting plates oppositely charged and separated by 2.3 cm. What is the surface charge density assumed uniform on plates.

Sol.

DATA

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\sigma = ?$$

CALCULATIONS

To keep electron in stationary its weight mg must be balanced by force qE . The force is directed upward.

$$F_g = F_e$$

$$mg = qE$$

$$E = \frac{mg}{q}$$

The electric field between two oppositely charged plates is

$$E = \frac{\sigma}{\epsilon_0}$$

$$\sigma = \epsilon_0 E$$

$$\sigma = \frac{mg}{q} \epsilon_0$$

$$\sigma = \frac{9.1 \times 10^{-31} \times 9.8 \times 8.85 \times 10^{-12}}{1.6 \times 10^{-19}}$$

$$\sigma = 4.9 \times 10^{-22} \text{ C/m}^2$$

- P.16** A very long straight thin wire carries -3.6 nC/m of fixed negative charge. The wire is to be surrounded by a uniform cylinder of positive charge, and radius 1.5 cm , coaxial with the wire. The volume charge density of the cylinder to be selected so that net electric field outside the cylinder is zero. Calculate required positive charge density ρ .

Sol.

DATA

$$\frac{q}{L} = 3.60 \times 10^{-9} \text{ C/m}$$

$$r = 1.50 \text{ cm}$$

$$\rho = ?$$

CALCULATIONS

$$\rho = \frac{q}{\text{vol}} = \frac{q}{\pi r^2 L}$$

$$\rho = \frac{3.60 \times 10^{-9}}{\pi (1.5 \times 10^{-2})^2}$$

$$\rho = 5.09 \times 10^{-6} \text{ C/m}^3$$

P.17 A 115 keV electron is fired directly towards a large flat plastic sheet that has a surface charge density $-2.08 \mu\text{C}/\text{m}^2$. From what distance must the electron be fired if it is just to fail to strike the sheet?

Sol.

DATA

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\text{KE} = 115 \times 10^3 \times 1.6 \times 10^{-19} \text{ J} = 1.84 \times 10^{-14} \text{ J}$$

$$\sigma = -2.08 \times 10^{-6} \text{ C}/\text{m}^2$$

$$d = ?$$

CALCULATIONS

$$F d = \text{KE}$$

$$q E d = \text{KE}$$

$$q \left(\frac{\sigma}{2\epsilon_0} \right) d = 1.84 \times 10^{-14}$$

$$d = \frac{2 \times 8.85 \times 10^{-12} \times 1.84 \times 10^{-14}}{1.6 \times 10^{-19} \times 2.08 \times 10^{-6}}$$

$$d = 0.97 \text{ m}$$

